

Table 2: Evaluation results of 32 SOTA models and 2 models SFT on Spatial-DISE. Row colors: Base, Δ vs base, Reasoning, Spatial, SFT on Spatial-DISE-12k. A Δ row shows the *absolute change in percentage points (pp)* relative to its base model and is placed between the parent and the derived model. Values are accuracy (%); brackets use [lower, upper] for the 95% CI. **Bold** indicates the highest accuracy; Underline indicates the second highest.

Model Tree		Acc. [95% CI]	E-D [95% CI]	E-S [95% CI]	I-D [95% CI]	I-S [95% CI]
Proprietary Bases						
Claude 3.7 Sonnet	Base	30.6% [26.8, 34.3]	22.6% [14.3, 32.1]	31.4% [21.4, 42.9]	32.4% [27.4, 37.7]	31.0% [21.8, 41.4]
Doubao1.5VL	Base	33.8% [29.9, 37.7]	31.0% [21.4, 40.5]	<u>37.1%</u> [25.7, 48.6]	33.6% [28.6, 39.0]	34.5% [25.3, 44.8]
		($\uparrow 8.2$)	($\uparrow 30.9$)	($\downarrow 5.7$)	($\uparrow 7.3$)	($\uparrow 1.1$)
- Doubao1.5VL-thinking	RLHF+RLVF	42.0% [37.9, 46.2]	61.9% [51.2, 72.6]	31.4% [21.4, 42.9]	40.9% [35.5, 46.2]	35.6% [25.3, 46.0]
Gemini 2.0 Flash	Base	34.2% [30.4, 37.9]	25.0% [15.5, 34.5]	31.4% [21.4, 42.9]	41.8% [36.5, 47.2]	17.2% [9.2, 25.3]
Gemini 2.5 Flash	Base	31.5% [27.7, 35.2]	16.7% [9.5, 25.0]	27.1% [17.1, 37.1]	39.3% [33.9, 44.7]	20.7% [12.6, 29.9]
Gemini 2.5 Flash w/o thinking	Base	32.0% [28.3, 35.8]	15.5% [8.3, 23.8]	28.6% [18.6, 38.6]	39.6% [34.3, 45.0]	23.0% [14.9, 32.2]
GPT-4.1 nano	Base	29.3% [25.6, 33.1]	29.8% [20.2, 40.5]	35.7% [25.7, 47.1]	31.1% [26.1, 36.2]	17.2% [9.2, 25.3]
GPT-4o	Base	28.1% [24.5, 31.8]	26.2% [16.7, 35.7]	22.9% [12.9, 32.9]	29.9% [24.8, 34.9]	27.6% [18.4, 36.8]
GPT-4o-mini	Base	25.6% [22.0, 29.2]	16.7% [9.5, 25.0]	21.4% [12.8, 31.4]	28.0% [23.0, 33.0]	28.7% [19.5, 37.9]
GPT-5	Base	30.1% [26.3, 34.0]	23.8% [15.5, 33.3]	25.7% [15.7, 35.7]	33.6% [28.6, 39.0]	26.4% [17.2, 35.6]
o4-mini	Base	33.3% [29.5, 37.2]	16.7% [9.5, 25.0]	25.7% [15.7, 35.7]	36.8% [31.8, 42.1]	42.5% [32.2, 52.9]
Proprietary Average		31.9% [29.0, 34.7]	26.0% [17.2, 34.8]	28.9% [25.6, 32.3]	35.2% [32.0, 38.4]	27.7% [22.3, 33.0]
Open-source Bases						
Llama3V-11B	Base	24.5% [20.9, 28.1]	29.8% [20.2, 39.3]	14.3% [7.1, 22.9]	25.5% [20.8, 30.5]	24.1% [14.9, 33.3]
		($\downarrow 0.5$)	(-)	($\uparrow 8.6$)	($\downarrow 1.0$)	($\downarrow 6.9$)
- LLaVA-CoT	CoT	24.0% [20.6, 27.5]	29.8% [20.2, 39.3]	22.9% [12.9, 32.9]	24.5% [19.8, 29.2]	17.2% [10.3, 25.3]
Cambrian-13B	Base	26.7% [23.1, 30.4]	25.0% [16.7, 34.5]	32.9% [21.4, 44.3]	25.8% [21.1, 30.8]	26.4% [17.2, 35.6]
Cambrian-8B	Base	22.9% [19.5, 26.3]	19.0% [10.7, 27.4]	15.7% [7.1, 24.3]	23.9% [19.2, 28.6]	28.7% [19.5, 37.9]
InternVL3-38B	Base	32.4% [28.6, 36.3]	27.4% [17.9, 36.9]	30.0% [20.0, 41.4]	35.8% [30.8, 41.2]	26.4% [17.2, 35.6]
InternVL3-14B	Base	31.1% [27.4, 34.9]	21.4% [13.1, 29.8]	31.4% [20.0, 42.9]	37.1% [31.8, 42.5]	18.4% [10.3, 26.4]
InternVL3-8B	Base	26.3% [22.7, 29.9]	23.8% [15.5, 33.3]	28.6% [18.6, 40.0]	30.8% [25.8, 35.8]	10.3% [4.6, 17.2]
Kimi-VL-A3B	Base	24.3% [20.8, 27.9]	17.9% [9.5, 26.2]	27.1% [17.1, 37.1]	27.7% [22.6, 32.7]	16.1% [9.2, 24.1]
		($\uparrow 0.4$)	($\uparrow 9.5$)	($\uparrow 1.5$)	($\downarrow 3.8$)	($\uparrow 5.7$)
- Kimi-VL-Thinking	CoT+RL	24.7% [21.1, 28.3]	27.4% [17.9, 36.9]	28.6% [18.6, 38.6]	23.9% [19.2, 28.6]	21.8% [13.8, 31.0]
Ovis2-16B	Base	26.3% [22.7, 29.9]	20.2% [11.9, 28.6]	27.1% [17.1, 38.6]	31.4% [26.4, 36.8]	12.6% [5.7, 19.5]
Ovis2-8B	Base	23.8% [20.4, 27.4]	15.5% [8.3, 23.8]	21.4% [12.9, 31.4]	29.6% [24.5, 34.6]	12.6% [5.7, 20.7]
Qwen2.5-VL-32B	Base	27.2% [23.4, 30.9]	21.4% [13.1, 29.8]	31.4% [21.4, 42.9]	27.7% [23.0, 32.7]	27.6% [18.4, 37.9]
Qwen2.5-VL-7B	Base	26.1% [22.5, 29.9]	32.1% [22.6, 42.9]	24.3% [14.3, 34.3]	27.7% [22.6, 32.7]	16.1% [9.2, 24.1]
		($\uparrow 1.8$)	($\downarrow 4.7$)	($\uparrow 2.8$)	($\uparrow 0.9$)	($\uparrow 10.3$)
- VLAA-Thinker-7B	GRPO	27.9% [24.3, 31.7]	27.4% [17.9, 36.9]	27.1% [17.1, 37.1]	28.6% [23.9, 33.6]	26.4% [17.2, 35.6]
		($\uparrow 20.9$)	($\uparrow 34.6$)	($\uparrow 11.4$)	($\uparrow 15.4$)	($\uparrow 35.6$)
- Qwen2.5-VL-7B-sft	SFT (SD-12k)	47.0% [42.9, 51.2]	66.7% [56.0, 76.2]	35.7% [24.3, 47.1]	43.1% [37.7, 48.7]	51.7% [41.4, 62.1]
		($\uparrow 0.9$)	($\downarrow 2.3$)	($\downarrow 7.2$)	($\uparrow 1.9$)	($\uparrow 6.9$)
- SpaceR	SG-RLVR	27.0% [23.4, 30.8]	29.8% [20.2, 39.3]	17.1% [8.6, 27.1]	29.6% [24.5, 34.6]	23.0% [14.9, 32.2]
Qwen2.5-VL-3B	Base	22.9% [19.5, 26.5]	25.0% [15.5, 34.5]	17.1% [8.6, 25.7]	26.4% [21.7, 31.4]	12.6% [5.7, 20.7]
		($\uparrow 3.2$)	($\uparrow 4.8$)	($\uparrow 2.9$)	(-)	($\uparrow 13.8$)
- LMM-R1	PPO	26.1% [22.5, 29.9]	29.8% [20.2, 39.3]	20.0% [11.4, 30.0]	26.4% [21.7, 31.4]	26.4% [17.2, 35.6]
		($\uparrow 7.9$)	($\uparrow 8.3$)	($\uparrow 1.5$)	($\uparrow 7.2$)	($\uparrow 15.0$)
- VLM-R1	GRPO	30.8% [27.0, 34.7]	33.3% [23.8, 44.0]	18.6% [10.0, 28.6]	33.6% [28.6, 39.0]	27.6% [18.4, 36.8]
		($\uparrow 3.0$)	($\uparrow 3.6$)	($\uparrow 12.9$)	($\uparrow 1.3$)	($\uparrow 1.2$)
- VLAA-Thinker-3B	GRPO	25.9% [22.4, 29.5]	28.6% [19.0, 38.1]	30.0% [20.0, 41.4]	27.7% [23.0, 32.7]	13.8% [6.9, 21.8]
		($\uparrow 6.7$)	($\uparrow 10.7$)	($\downarrow 7.1$)	($\uparrow 7.2$)	($\uparrow 11.5$)
- SpaceThinker	SFT	32.6% [25.4, 32.9]	39.3% [20.2, 40.5]	22.9% [15.7, 35.7]	34.9% [27.7, 37.7]	25.3% [10.3, 26.4]
			($\uparrow 2.4$)	($\downarrow 5.7$)	($\downarrow 1.0$)	($\uparrow 5.7$)
- SpaceOM	SFT	25.9% [22.4, 29.5]	31.0% [20.2, 39.3]	24.3% [14.3, 34.3]	26.7% [22.0, 31.8]	19.5% [11.5, 28.7]
		($\uparrow 15.4$)	($\uparrow 21.4$)	($\uparrow 2.8$)	($\uparrow 11.0$)	($\uparrow 35.7$)
- SpaceOM-sft	SFT (SD-12k)	41.3% [37.4, 45.4]	52.4% [41.7, 63.1]	27.1% [17.1, 37.1]	37.7% [32.4, 43.1]	55.2% [44.8, 65.5]
Open-source Average		26.2% [25.2, 27.3]	23.2% [20.7, 25.8]	25.1% [22.2, 28.0]	29.1% [27.7, 30.6]	19.3% [17.0, 21.7]
Human Level		76.8% [74.8, 78.9]	61.1% [56.6, 65.5]	81.1% [76.7, 85.5]	80.2% [78.2, 82.3]	76.8% [72.9, 80.7]
Random Guessing		24.8%	25.4%	26.3%	24.3%	24.7%

not learning spatial reasoning in a human-like, scaffolded manner. Instead of building dynamic capabilities upon a solid foundation of static scene understanding, they appear to be learning fragmented strategies, recognizing patterns of "change" without a robust, underlying model of the static world.

Computational rigor can outperform fallible human simulation. This is evident in Doubao-1.5-thinking model, which surpassed the human baseline on E-D tasks. This superior performance can likely be attributed to the nature of 2D/3D combination. As confirmed by our human performance analysis (Appendix, Table 9), these tasks are particularly arduous and cognitively demanding for humans. 3D Combination commanding the longest mean response time (59.2s) among all tasks for humans, who must rely on fallible mental simulation. In contrast, we observed that the Doubao model transforms these challenges into computational problems by employing a more algorithmic strategy to compute and compare geometric features of components—such as edges, angles, and connection points. Essentially, the model excels by converting a cognitively exhausting simulation task into a precise computational problem, a domain where it holds a distinct advantage over human intuition.

4.3 FINE-TUNING ON SPATIAL-DISE-12K

To ascertain whether Spatial-DISE-12K effectively enhances the spatial reasoning capabilities of vision-language models, and to evaluate the cross-domain generalizability of this training, we conducted supervised fine-tuning experiments. Specifically, two representative open-source VLMs—Qwen2.5-VL-7B and SpaceOm—were fine-tuned using Low-Rank Adaptation (LoRA) applied to all linear layers utilizing the Spatial-DISE-12K training split (see Appendix B.4 for implementation details). Following optimization, we structure our comprehensive evaluation around four core research questions (RQs), assessing the models on the Spatial-DISE Bench alongside five established external benchmarks: CVBench, SAT, SPACE, OmniSpatial, and VSIBench_MCQ.

Table 3: Accuracy for Qwen2.5-VL (Base vs SFT) and SpaceOm (Base vs SFT). Δ is SFT-Base in percentage points (pp).

	Spatial-DISE	CVBench	SAT	SPACE	OmniSpatial	VSIBench_MCQ
SpaceOm	25.9%	68.8%	46.67%	27.22%	27.91%	31.05%
+DISE SFT	41.3%	70.33%	49.33%	32.6%	34.28%	33.7%
Δ	($\uparrow 15.4\%$)	($\uparrow 1.53\%$)	($\uparrow 2.66\%$)	($\uparrow 5.38\%$)	($\uparrow 6.37\%$)	($\uparrow 2.65\%$)
Qwen2.5-VL-7B	26.1%	75.9%	65.3%	28.7%	21.8%	19.3%
+DISE SFT	47.0%	77.4%	69.3%	32.2%	34.0%	22.6%
Δ	($\uparrow 20.9\%$)	($\uparrow 1.5\%$)	($\uparrow 4.0\%$)	($\uparrow 3.5\%$)	($\uparrow 12.2\%$)	($\uparrow 3.3\%$)

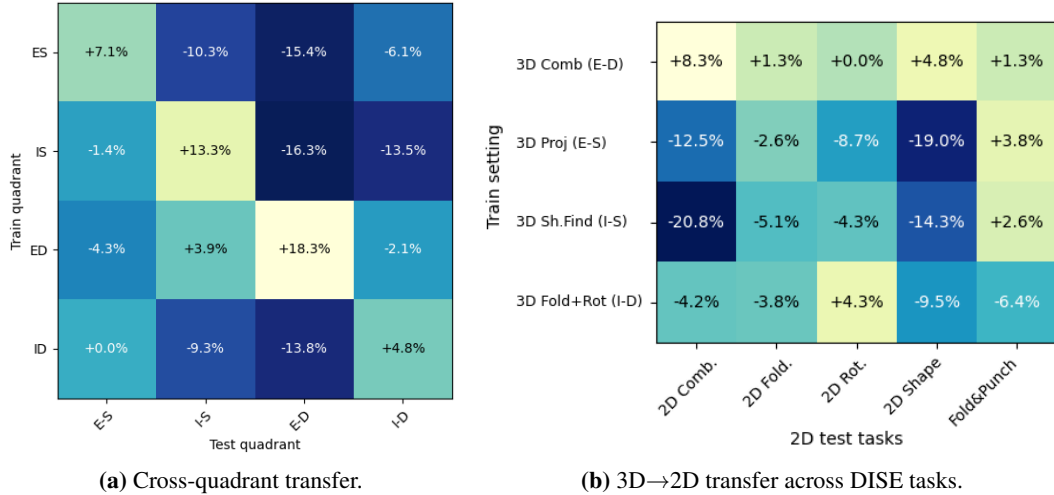


Figure 3: Transfer heatmap by fine-tuning Qwen2.5-VL-7B on Spatial-DISE quadrant-wise.

RQ1: Does 3D spatial training cultivate a broad, structured set of in-domain skills? To address this, we evaluated the fine-tuned models on the Spatial-DISE Bench. Empirical evaluation demonstrates that fine-tuning on Spatial-DISE-12K yields substantial performance gains. Specifically, the overall accuracy of Qwen2.5-VL-7B improved from 26.1% to 47.0%, and SpaceOm from 25.9% to 41.3%. The most pronounced enhancements were observed in Intrinsic-Dynamic and Extrinsic-Dynamic tasks, alongside a remarkable surge in Qwen2.5-VL-7B’s Intrinsic-Static performance (scaling from 16.1% to 51.7%).

RQ2: Do models learn factorized spatial dimensions or entangled, quadrant-specific representations? To elucidate the underlying mechanisms driving these gains, we isolated the training curriculum by fine-tuning Qwen2.5-VL-7B exclusively on individual DISE quadrants and visualized the subsequent accuracy fluctuations (Figure 3a). The resulting heatmap reveals a robust diagonal pattern, indicating profound quadrant-specific specialization: training on a specific quadrant predominantly amplifies performance within that corresponding domain. Conversely, off-diagonal transfer is largely minimal or even detrimental, failing to exhibit clean factorization along the Intrinsic/Extrinsic or Static/Dynamic axes. We observe distinct directional asymmetries (e.g., a mild positive transfer of +3.9 pp from 3D E-D to I-S, which starkly contrasts with the strongly negative transfer from I-S to E-D). This implies that the four DISE quadrants encapsulate relatively distinct cohorts of spatial

competencies; strengthening one family does not automatically confer a “universal” spatial skill, and overall benchmark gains necessitate exposure to multiple 3D quadrants during training.

RQ3: Can 3D spatial training induce representations that transfer to 2D reasoning tasks? To probe cross-dimensional transferability, we maintained the single-quadrant fine-tuning paradigm but evaluated the resultant models across all 2D DISE tasks (Figure 3b). The findings expose qualitatively divergent reasoning regimes. Training on extrinsic-dynamic 3D tasks catalyzed broadly positive transfer across varied 2D settings, suggesting that scene-centric dynamic reasoning fosters highly reusable representations applicable to 2D projection, combination, and occlusion problems. In stark contrast, training on intrinsic-static or extrinsic-static 3D tasks yielded narrowly constrained or even negative transfer. Furthermore, while intrinsic-dynamic training facilitated 2D rotation performance, it concurrently degraded the model’s capacity to solve simpler 2D problems. These asymmetric patterns underscore that object-centric static reasoning tends to form overly specialized schemas, which can actively interfere with tasks reliant on relative or dynamic reference frames.

RQ4: Do the acquired spatial competencies generalize to external benchmarks without catastrophic forgetting? To assess broader applicability, we evaluated the fine-tuned models on a suite of external spatial benchmarks (Table 3). Fine-tuning on Spatial-DISE engendered consistent yet highly selective out-of-domain gains, importantly demonstrating that this targeted spatial curriculum does not induce catastrophic forgetting of general visual capabilities (e.g., Qwen2.5-VL-7B-sft improved marginally on CVBench). The most substantial improvements materialized on the SPACE and OmniSpatial benchmarks, which intrinsically emphasize viewpoint transformations and 3D-consistent reasoning. Conversely, benchmarks amalgamating spatial reasoning with generalized language or diagrammatic comprehension exhibited more modest benefits. This corroborates our prior analyses: Spatial-DISE-12K fortifies specific spatial reasoning architectures that subsequently transfer primarily to structurally homologous external tasks.

Ultimately, even following rigorous fine-tuning, the apex model (Qwen2.5-VL-7B-sft) remains markedly inferior to the human baseline on the Spatial-DISE Bench, highlighting substantial avenues for future architectural refinement. Collectively, the responses to these research questions demonstrate that while contemporary VLMs do not yet possess robust, human-like spatial schemas, meticulously structured 3D training regimens like Spatial-DISE-12K can systematically cultivate distinct spatial skill cohorts and induce meaningful, albeit selective, cross-task and cross-benchmark transfer.

5 ERROR ANALYSIS

5.1 ERROR TAXONOMY AND MAIN DEFICITS

Given the persistent performance gap between fine-tuned VLMs and human baselines, we must move beyond simply measuring what models fail at to understanding why they fail. To provide this cognitive diagnosis, we employ Doubao-1.6-thinking as an automated judge and, combined with human analysis, analyze a sample of 200 incorrect responses from four representative models: **GeminiFlash2-0**, **Qwen2.5-VL-3B**, **Doubao-1.5-thinking**, and **SpaceThinker**, with 50 samples drawn from each.

We established a high-level error taxonomy to systematically diagnose failures by deconstructing the model mistakes into three errors: **Perceptual Error**, **Comprehension Error** and **Reasoning Error**.

The analysis reveals that Reasoning errors are the predominant failure category, accounting for an overwhelming 72.5% of all analyzed failure responses. Perceptual errors constituted 17.5% of the total, while comprehension errors were the least common at 10%. This distribution strongly suggests that the primary bottleneck for current VLMs is not in visual perception but in complex spatial-logical inference. The predominance of reasoning errors (145) prompted a deeper analysis, which identified three fundamental cognitive deficits.

Table 4: Error Types and Their Frequencies.

Major Error	Sub-category	Num.
Reasoning Err.	Failure in Rule Application	65
	Failure in Mental Simulation	58
	Failure in Holistic-Local Processing	22
Perceptual Err.	—	35
Comprehension Err.	—	20